

# Area of Trapezoids

## Lesson 5-2

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

### Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*A table top wider than the bottom shelf — the top and bottom edges are parallel, the two sides slant inward*

#### Trapezoid

A four-sided shape with just one pair of parallel sides.

*If the bottom of the trapezoid is 10 ft, then  $b_1 = 10$  ft in the formula  $A = \frac{1}{2}(b_1 + b_2) \times h$*

#### Base 1 ( $b_1$ )

One of the two parallel sides of a trapezoid.

*If the top of the trapezoid is 6 ft, then  $b_2 = 6$  ft; both bases are parallel to each other*

#### Base 2 ( $b_2$ )

The other parallel side of a trapezoid.

*A dashed vertical line from the top base straight down to the bottom base at a  $90^\circ$  angle — NOT the slanted side*

#### Height

The straight-up distance between the two parallel sides.

*A trapezoid with  $b_1 = 10$ ,  $b_2 = 6$ ,  $h = 4$  has area =  $\frac{1}{2}(10 + 6) \times 4 = 32$  sq units*

#### Area

How much space is inside a flat shape.

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

I can find the area of a trapezoid using the formula  $A = \frac{1}{2}(b_1 + b_2) \times h$ .

#### 1 Notice

Your architecture firm is designing a trapezoidal window for a modern building.

#### 2 Key idea

I can find the area of a trapezoid using the formula  $A = \frac{1}{2}(b_1 + b_2) \times h$ .

#### 3 Words I'll use

Trapezoid

Base 1 ( $b_1$ )

Base 2 ( $b_2$ )

Height

Area



#### Think About It

- What shape is the window?
- How many parallel sides does it have?
- What measurements are given?



#### Watch

Watch your teacher model one example. Jot what you see.

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#### We try

Solve the next one together as a class.

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#### You try

Now try one on your own.

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**The trapezoidal window has a top edge of 4 feet and a bottom edge of 8 feet. Why does the area formula use BOTH bases instead of just one?**

#### Level 1 support

**Start here:** We use both bases because the top and bottom are different lengths.

 **Need a hint?**

**Sentence starters (optional)**

**We use both bases because the top and bottom are \_\_\_\_.**

*Usamos las dos bases porque la parte de arriba y la de abajo son \_\_\_\_.*

**If I used only one base, the area would be \_\_\_\_.**

*Si usara solo una base, el área sería \_\_\_\_.*

**Word bank:** trapezoid base 1 (b1) base 2 (b2) height area

**Listen for:** Students explain a trapezoid has two parallel sides of different lengths, so using only one would over- or under-count the area; strong answers mention averaging the bases.

#### Level 2

**The wonder prompt asks what happens if both bases are equal. Predict the shape and explain how the trapezoid formula would simplify in that case.**

- If b1 equals b2, the shape becomes a \_\_\_\_ because \_\_\_\_.
- The formula simplifies to \_\_\_\_, since \_\_\_\_.

**EXPLORE**

On the grid you copied the trapezoid (bases 8 and 4, height 5), flipped it, and joined it to the original. What shape formed, and how does that explain the  $\frac{1}{2}$  in the formula?

**Level 1 support**

**Start here:** Two trapezoids make a parallelogram, so one trapezoid is half of it.

 **Need a hint?**

**Sentence starters (optional)**

**When I flip and join the trapezoid, I make a \_\_\_\_.**

*Cuando volteo y uno el trapecio, formo un \_\_\_\_.*

**I added the two bases first because \_\_\_\_.**

*Primero sumé las dos bases porque \_\_\_\_.*

**The area is  $\frac{1}{2}$  times (8 plus 4) times 5, which equals \_\_\_\_.**

*El área es  $\frac{1}{2}$  por (8 más 4) por 5, que es \_\_\_\_.*

**Word bank:** **trapezoid** **parallelogram** **base 1 (b1)** **base 2 (b2)** **height**

**Listen for:** A strong answer names a parallelogram, links the  $\frac{1}{2}$  to taking half of two combined trapezoids, and computes  $\text{area} = \frac{1}{2} \times 12 \times 5 = 30 \text{ sq ft}$ .

**Level 2**

**Compare finding a trapezoid's area to averaging the two bases and multiplying by the height. Are these the same method? Justify with the window numbers.**

- Averaging the bases gives \_\_\_\_, and multiplying by the height gives \_\_\_\_, which \_\_\_\_ the formula answer.
- These methods are \_\_\_\_ because \_\_\_\_.

CONNECT

The builder pours concrete for a trapezoidal driveway with edges of 20 ft and 12 ft and a height of 15 ft; one bag covers 6 sq ft. Walk through finding the number of bags.

Level 1 support

**Start here:** Find the trapezoid area first, then divide by what one bag covers.

 **Need a hint?**

**Sentence starters (optional)**

**The driveway area is  $\frac{1}{2}$  times (20 plus 12) times 15, which equals \_\_\_\_ square feet.**

*El área de la entrada es  $\frac{1}{2}$  por (20 más 12) por 15, que es \_\_\_\_ pies cuadrados.*

**He needs \_\_\_\_ bags because \_\_\_\_ divided by 6 equals \_\_\_\_.**

*Necesita \_\_\_\_ bolsas porque \_\_\_\_ entre 6 es \_\_\_\_.*

**Word bank:** trapezoid base 1 (b1) base 2 (b2) height area

**Listen for:** Students compute area = 240 sq ft, then  $240 / 6 = 40$  bags, and explain why dividing by coverage gives the bag count.

Level 2

**If the builder doubled only the height of the driveway, would the number of bags double? Justify using the formula.**

- Doubling the height \_\_\_\_ the area because \_\_\_\_.
- So the number of bags would \_\_\_\_, since \_\_\_\_.

## Guided Examples

### Example 1

**What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?**

**Solution:**  $A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (6 + 10) \times 4 = \frac{1}{2} \times 16 \times 4 = 32$  sq cm.

**Answer:** A. 32 sq cm

### Example 2

**A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?**

**Solution:**  $A = \frac{1}{2} \times (b_1 + b_2) \times h \rightarrow 45 = \frac{1}{2} \times 18 \times h \rightarrow 45 = 9h \rightarrow h = 5$  ft.

**Answer:** A. 5 ft

### Example 3

**What is the first step when finding the area of a trapezoid?**

**Solution:** The formula is  $A = \frac{1}{2} \times (b_1 + b_2) \times h$ . The first step is to add the two parallel bases.

**Answer:** A. Add the two bases together

## Write About the Math

### The Writing Revolution

I can explain my steps using the words trapezoid, base, height, and area.

#### 1. Kernel Sentence subject + verb

**Model:** Trapezoid is a four-sided shape with just one pair of parallel sides.

*Trapecio es una figura de cuatro lados con solo un par de lados paralelos.*

**Write a kernel sentence about trapezoid. Use a subject and a verb.**

*Escribe una oración base sobre trapecio. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because • but • so

**Kernel:** Trapezoid matters in math

*Trapecio importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Trapezoid matters in math because \_\_\_\_.**

*Trapecio importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Trapezoid matters in math, but \_\_\_\_.**

*Trapecio importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Trapezoid matters in math, so \_\_\_\_.**

*Trapecio importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

#### Statement *Afirmación*

Tell one true fact about trapezoid.  
*Di un hecho verdadero sobre trapezoid.*

**Trapezoid \_\_\_\_.**

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#### Question *Pregunta*

Ask a question about trapezoid.  
*Haz una pregunta sobre trapezoid.*

**How does \_\_\_\_ ?**

*¿Cómo \_\_\_\_ ?*

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#### Exclamation *Exclamación*

Show excitement about trapezoid.  
*Muestra entusiasmo sobre trapezoid.*

**Wow, \_\_\_\_ !**

*¡Guau, \_\_\_\_ !*

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#### Command *Mandato*

Tell a partner what to do with trapezoid.  
*Dile a un compañero qué hacer con trapezoid.*

**First, \_\_\_\_ .**

*Primero, \_\_\_\_ .*

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### 4. Explain Your Reasoning use a sentence starter

**We use both bases because the top and bottom are \_\_\_\_.**

*Usamos las dos bases porque la parte de arriba y la de abajo son \_\_\_\_.*

**If I used only one base, the area would be \_\_\_\_.**

*Si usara solo una base, el área sería \_\_\_\_.*

**When I flip and join the trapezoid, I make a \_\_\_\_.**

*Cuando volteo y uno el trapecio, formo un \_\_\_\_.*

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## Try It

Solve on your own. Check the answer key when you are done.

**1. What is the first step when finding the area of a trapezoid?**

- A. Add the two bases together
- B. Multiply the two bases
- C. Subtract the shorter base from the longer base
- D. Divide the height by 2

Show your work:

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**2. A trapezoid has bases of 5 in and 9 in, and a height of 6 in. What is its area?**

- A. 42 sq in
- B. 84 sq in
- C. 27 sq in
- D. 30 sq in

Show your work:

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### Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A trapezoid and a parallelogram both have a height of 8 ft. The trapezoid has bases of 5 ft and 11 ft. The parallelogram has a base of 8 ft. Which shape has the greater area? Explain your reasoning using the formulas.**

*Sentence starter: The trapezoid's area is \_\_\_\_ because  $\frac{1}{2} \times (\text{ } + \text{ }) \times \text{ } = \text{ }$ . The parallelogram's area is \_\_\_\_ because  $\text{ } \times \text{ } = \text{ }$ . The \_\_\_\_ has a greater area by \_\_\_\_ sq ft.*

Show your work:

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### Reflect — Exit Ticket

**A trapezoid has bases of 9 inches and 5 inches, and a height of 6 inches. What is its area?**

- A. 42 sq in
- B. 84 sq in
- C. 27 sq in
- D. 42 in

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** A. Add the two bases together — *The formula is  $A = \frac{1}{2} \times (b_1 + b_2) \times h$ . The first step is to add the two parallel bases.*
2. **Try It 2:** A. 42 sq in —  $A = \frac{1}{2} \times (5 + 9) \times 6 = \frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42 \text{ sq in.}$
3. **Exit Ticket:** A. 42 sq in —  $A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (9 + 5) \times 6 = \frac{1}{2} \times 14 \times 6 = 42 \text{ square inches.}$

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Trapezoid is a four-sided shape with just one pair of parallel sides.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).